VEGA

version 1.2.5

USER'S GUIDE

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1.Introduction

VEGA is an application that allows to apply several QSAR models to the given chemicals provided by the user, returning as output complete reports (in plain text and in PDF format) containing the predictions and all needed supporting information. It is a JAVA standalone application and it does not require any network communication (chemicals are processed on the local machine, no information is shared across the Internet). Please note that an internet connection is anyway required at its first run to download and install some needed additional files.

Currently, there are **142 models** implemented in VEGA:

Model Section	Endpoint	Model
Human Toxicity	Mutagenicity (Ames test)	Mutagenicity (Ames test) model (CAESAR)
Human Toxicity	Mutagenicity (Ames test)	Mutagenicity (Ames test) model (ISS)
Human Toxicity	Mutagenicity (Ames test)	Mutagenicity (Ames test) model (SarPy-IRFMN)
Human Toxicity	Mutagenicity (Ames test)	Mutagenicity (Ames test) model (KNN-Read-Across)
Human Toxicity	Mutagenicity (Ames test)	Mutagenicity (Ames test) model for aromatic amines (CONCERT/IRFMN)
Human Toxicity	Mutagenicity (Ames test)	Mutagenicity (Ames test) CONSENSUS model
Human Toxicity	Developmental toxicity	Developmental Toxicity model (CAESAR)
Human Toxicity	Developmental toxicity	Developmental/Reproductive Toxicity library (PG)
Human Toxicity	Carcinogenicity	Carcinogenicity model (CAESAR)
Human Toxicity	Carcinogenicity	Carcinogenicity model (ISS)
Human Toxicity	Carcinogenicity	Carcinogenicity model (IRFMN-ISSCAN-CGX)
Human Toxicity	Carcinogenicity	Carcinogenicity model (IRFMN-Antares)
Human Toxicity	Carcinogenicity	Carcinogenicity oral classification model (IRFMN)
Human Toxicity	Carcinogenicity	Carcinogenicity oral Slope Factor model (IRFMN)
Human Toxicity	Carcinogenicity	Carcinogenicity inhalation classification model (IRFMN)
Human Toxicity	Carcinogenicity	Carcinogenicity inhalation Slope Factor model (IRFMN)
Human Toxicity	Carcinogenicity	Carcinogenicity in male rat (CORAL)
Human Toxicity	Carcinogenicity	Carcinogenicity in female Rat (CORAL)
Human Toxicity	Acute Toxicity (LD50)	Acute Toxicity (LD50) model (KNN)
Human Toxicity	Skin Sensitization	Skin Sensitization model (CAESAR)
Human Toxicity	Skin Sensitization	Skin Sensitization model (IRFMN-JRC)
Human Toxicity	Skin Sensitization	Skin Sensitization model (NCSTOX)
Human Toxicity	Skin Sensitization	Skin Sensitization model (TOXTREE)
Human Toxicity	Skin Sensitization	Skin Sensitization (CONCERT/Kode)
Human Toxicity	Skin Sensitization	Skin Sensitization (CONCERT/SarPy)
Human Toxicity	Skin Irritation	Skin Irritation (CONCERT/Kode)
Human Toxicity	Skin Irritation	Skin Irritation (CONCERT/Coral)
Human Toxicity	Skin Irritation	Skin Irritation model (CONCERT/SarPy)
Human Toxicity	Eye Irritation	Eye Irritation (CONCERT/Kode)

Human Toxicity	Eye Irritation	Eye Irritation (CONCERT/KNN)
Human Toxicity	Eye Irritation	Eye Irritation (CONCERT/SarPy)
Human Toxicity	Chromosomal aberration	Chromosomal aberration model (CORAL)
Human Toxicity	Micronucleus assay	In vitro Micronucleus activity (IRFMN-VERMEER)
Human Toxicity	Micronucleus assay	In vivo Micronucleus activity (IRFMN)
Human Toxicity	Estrogen receptor effect	Estrogen Receptor-mediated effect (IRFMN-CERAPP)
Human Toxicity	Estrogen receptor effect	Estrogen Receptor Relative Binding Affinity model (IRFMN)
Human Toxicity	Androgen receptor effect	Androgen Receptor-mediated effect (IRFMN-COMPARA)
Human Toxicity	Thyroid receptor effect	Thyroid Receptor Alpha effect (NRMEA)
Human Toxicity	Thyroid receptor effect	Thyroid Receptor Beta effect (NRMEA)
Human Toxicity	Glucocorticoid Receptor effect	Glucocorticoid Receptor (OBERON)
Human Toxicity	Thyroperoxidase Inhibitory activity	Thyroperoxidase Inhibitory Activity (OBERON)
Human Toxicity	Steroidogenesis activity	Steroidogenesis (OBERON)
Human Toxicity	Endocrine Disruptor activity	Endocrine Disruptor activity screening (IRFMN)
Human Toxicity	NOAEL	NOAEL (CONCERT/Coral)
Human Toxicity	NOAEL	NOAEL (IRFMN-CORAL)
Human Toxicity	NOAEL	Liver NOAEL (CORAL)
Human Toxicity	LOAEL	LOAEL (CONCERT/Coral)
Human Toxicity	LOAEL	Liver LOAEL (CORAL)
Human Toxicity	Cramer classification	Cramer classification (TOXTREE)
Human Toxicity	Hepatotoxicity	Hepatotoxicity model (IRFMN)
Human Toxicity	Hepatotoxicity	Drug-Induced Liver Injury (Bayer/ONTOX)
Human Toxicity	Cardiotoxicity	Apical Cardio Tox
Human Toxicity	Cardiotoxicity	Cardio Tox Multitask
Human Toxicity	Mitochondrial dysfunction	Mitochondrial Dysfunction
Human Toxicity	Kidney failure (KF)	ACE - Angiotensin Converting Enzyme model (ONTOX)
Human Toxicity	Kidney failure (KF)	OAT1 - Organic Anion Transporter model (ONTOX)
Human Toxicity	Kidney failure (KF)	COX1 - Cyclooxygenase 1 model (ONTOX)
Human Toxicity	Kidney failure (KF)	AT1R - Angiotensin II Receptor type 1 model (ONTOX)
Human Toxicity	Cholestasis (CHO)	BSEP - Bile Salt Export Pump model (ONTOX)
Human Toxicity	Neural tube closure (NTD)	HDeac - Histone Deacetylase model (ONTOX)
Human Toxicity	Neural tube closure (NTD)	CYP26 - Cytochrome P450 26A1 model (ONTOX)
Human Toxicity	Neural tube closure (NTD)	BMP - Bone Morphogenetic Proteins model (ONTOX)
Human Toxicity	Neural tube closure (NTD)	WNT - Protein Wnt model (ONTOX)
Human Toxicity	Neural tube closure (NTD)	FGFR1 - Fibroblast Growth Factor Receptor isoform 1 model (ONTOX)
Human Toxicity	Neural tube closure (NTD)	FGFR2 - Fibroblast Growth Factor Receptor isoform 2 model (ONTOX)
Human Toxicity	Neural tube closure (NTD)	FGFR3 - Fibroblast Growth Factor Receptor isoform 3 model (ONTOX)
Human Toxicity	Neural tube closure (NTD)	FGFR4 - Fibroblast Growth Factor Receptor isoform 4 model (ONTOX)
Human Toxicity	Cognitive function defects (CFD)	TTR - Transthyretin model (ONTOX)

Human Toxicity	Cognitive function defects (CFD)	ACHE - Acetylcholinesterase model (ONTOX)
Human Toxicity	Cognitive function defects (CFD)	THRb - Thyroid Hormone Receptor beta model (ONTOX)
Human Toxicity	Cognitive function defects (CFD)	THRa - Thyroid Hormone Receptor alpha model (ONTOX)
Human Toxicity	Cognitive function defects (CFD)	VGSC - Voltage-Gated Sodium Channels model (ONTOX)
Human Toxicity	Cognitive function defects (CFD)	NMDA - N-Methyl-D-Aspartate receptor (ONTOX)
Human Toxicity	Cognitive function defects (CFD)	PPARd - Peroxisome Proliferator-Activated Receptor delta model (ONTOX)
Human Toxicity	Cognitive function defects (CFD)	GR - Glucocorticoid Receptor Model (ONTOX)
Human Toxicity	Liver steatosis (STE)	PXR - Pregnane X Receptor model (ONTOX)
Human Toxicity	Liver steatosis (STE)	AHR - Aryl Hydrocarbon Receptor model (ONTOX)
Human Toxicity	Liver steatosis (STE)	PPARa - Peroxisome Proliferator-Activated Receptor alpha model (ONTOX)
Human Toxicity	Liver steatosis (STE)	PPARg - Peroxisome Proliferator-Activated Receptor gamma model (ONTOX)
EcoToxicity	BCF	BCF model (CAESAR)
EcoToxicity	BCF	BCF model (Meylan)
EcoToxicity	BCF	BCF model (Arnot-Gobas)
EcoToxicity	BCF	BCF model (KNN-Read-Across)
EcoToxicity	Aquatic Acute Toxicity	Fish Acute (LC50) Toxicity model (IRFMN)
EcoToxicity	Aquatic Acute Toxicity	Fish Acute (LC50) Toxicity model (NIC)
EcoToxicity	Aquatic Acute Toxicity	Fish Acute (LC50) Toxicity model (KNN-Read-Across)
EcoToxicity	Aquatic Acute Toxicity	Fish Acute (LC50) Toxicity classification (SarPy-IRFMN)
EcoToxicity	Aquatic Acute Toxicity	Fish Acute (LC50) Toxicity model (IRFMN-Combase)
EcoToxicity	Aquatic Acute Toxicity	Fathead Minnow LC50 96h (EPA)
EcoToxicity	Aquatic Acute Toxicity	Fathead Minnow LC50 model (KNN-IRFMN)
EcoToxicity	Aquatic Acute Toxicity	Daphnia Magna Acute (EC50) Toxicity model (IRFMN)
EcoToxicity	Aquatic Acute Toxicity	Daphnia Magna LC50 48h (EPA)
EcoToxicity	Aquatic Acute Toxicity	Daphnia Magna LC50 48h (DEMETRA)
EcoToxicity	Aquatic Acute Toxicity	Daphnia Magna Acute (EC50) Toxicity model (IRFMN-Combase)
EcoToxicity	Aquatic Acute Toxicity	Guppy LC50 model (KNN-IRFMN)
EcoToxicity	Aquatic Acute Toxicity	Algae Acute (EC50) Toxicity model (IRFMN)
EcoToxicity	Aquatic Acute Toxicity	Algae Classification Toxicity model (ProtoQSAR-Combase)
EcoToxicity	Aquatic Acute Toxicity	Algae Acute (EC50) Toxicity model (ProtoQSAR-Combase)
EcoToxicity	Aquatic Chronic Toxicity	Fish Chronic (NOEC) Toxicity model (IRFMN)
EcoToxicity	Aquatic Chronic Toxicity	Daphnia Magna Chronic (NOEC) Toxicity model (IRFMN)
EcoToxicity	Aquatic Chronic Toxicity	Algae Chronic (NOEC) Toxicity model (IRFMN)
EcoToxicity	Mode of Action	Verhaar classification (TOXTREE)
EcoToxicity	Mode of Action	MOA fish toxicity classification (EPA T.E.S.T.)
EcoToxicity	Mode of Action	MOA pesticide classification (IRFMN)
EcoToxicity	Terrestrial Acute Toxicity	Bee acute toxicity model (KNN-IRFMN)
EcoToxicity	Terrestrial Acute Toxicity	Earthworm Toxicity (CONCERT)

EcoToxicity	Sludge Toxicity	Sludge Classification Toxicity model (ProtoQSAR-Combase)
EcoToxicity	Sludge Toxicity	Sludge (EC50) Toxicity model (ProtoQSAR-Combase)
EcoToxicity	Zebrafish embryo activity	Zebrafish embryo AC50 (IRFMN-CORAL)
Fate & Distribution	Ready biodegradability	Ready Biodegradability model (IRFMN)
Fate & Distribution	Persistence (sediment)	Persistence (sediment) model (IRFMN)
Fate & Distribution	Persistence (sediment)	Persistence (sediment) quantitative model (IRFMN)
Fate & Distribution	Persistence (soil)	Persistence (soil) model (IRFMN)
Fate & Distribution	Persistence (soil)	Persistence (soil) quantitative model (IRFMN)
Fate & Distribution	Persistence (water)	Persistence (water) model (IRFMN)
Fate & Distribution	Persistence (water)	Persistence (water) quantitative model (IRFMN)
Fate & Distribution	Persistence (air)	Air Half-Life (IRFMN-CORAL)
Physical-Chemical properties	Octanol/Water partition coefficient (logP)	LogP model (Meylan-Kowwin)
Physical-Chemical properties	Octanol/Water partition coefficient (logP)	LogP model (MLogP)
Physical-Chemical properties	Octanol/Water partition coefficient (logP)	LogP model (ALogP)
Physical-Chemical properties	Water solubility	Water solubility model (IRFMN)
Physical-Chemical properties	Vapour pressure	Vapour Pressure (CONCERT/Kode)
Physical-Chemical properties	Melting point	Melting Point (CONCERT/Kode)
Physical-Chemical properties	Melting point	Melting Point (CONCERT/KNN)
Physical-Chemical properties	Hydrolysis	Hydrolysis (IRFMN-CORAL)
Physical-Chemical properties	Henry's law constant	Henry's Law model (OPERA)
Physical-Chemical properties	Octanol/air partition coefficient (KOA)	KOA model (OPERA)
Physical-Chemical properties	Soil adsorption coefficient of organic compounds (KOC)	KOC model (OPERA)
Human PBPK	Plasma Protein Binding	Plasma Protein Binding - LogK (IRFMN)
Human PBPK	Plasma Protein Binding	Plasma Protein Binding - sqFU (CORAL)
Human PBPK	Aromatase activity	Aromatase activity model (IRFMN)
Human PBPK	Aromatase activity	Aromatase activity model (TOX21)
Human PBPK	P-Glycoprotein activity	P-Glycoprotein activity model (NIC)
Human PBPK	Hepatic Steatosis MIE	Hepatic Steatosis MIE assay for PXR up (TOXCAST)
Human PBPK	Hepatic Steatosis MIE	Hepatic Steatosis MIE assay for PPARg up (TOXCAST)
Human PBPK	Hepatic Steatosis MIE	Hepatic Steatosis MIE assay for PPARa up (TOXCAST)
Human PBPK	Hepatic Steatosis MIE	Hepatic Steatosis MIE assay for NRF2 (TOXCAST)
Human PBPK	Skin permeation (LogKp)	Skin Permeation (LogKp) model (Potts and Guy)
Human PBPK	Skin permeation (LogKp)	Skin Permeation (LogKp) model (Ten Berge)

Human PBPK	Adipose tissue-blood partition	Adipose tissue - blood model (INERIS)
Human PBPK	Body elimination half-life	Total body elimination half-life (QSARINS)
Ecological PBPK	kM/Half Life	kM/Half-Life model (Arnot-EpiSuite)

2.Installation

VEGA a JAVA application, using Python (and Conda) to run some of its QSAR models. The application runs on a local machine and does not send any information to external on-line services.

To run VEGA, JAVA version 17 or higher is required. You can download OpenJDK (open source) from https://jdk.java.net/ or visit the OpenJDK website for more information (https://openjdk.java.net/). Alternatively, you can download the installers for the Oracle version of Java (not open source but available for free) from https://www.oracle.com/it/java/technologies/downloads/.

At its first run, VEGA will download and install a local distribution of Miniconda (Python), and then proceed to download the needed libraries for the QSAR models based on Python. A network connection is needed for this procedure. Please note that:

- The local distribution of Miniconda will not interfere with other Conda or Python distribution available on your machine
- The binaries for Miniconda are downloaded from its official websites, and the Python libraries downloaded frome their official repositories
- Additional files for the Python QSAR models are downloaded from a private VEGA server
- These procedures will not send nor share any information from your local machine to remote servers

This procedure will use approximately 7 GB of additional space on your machine. Please note that, as a relevant amount of data will be downloaded, this procedure could require some time.

To unistall VEGA, please run Vega-uninstall-WIN.bat (on Windows platforms) or Vega-uninstall-LINUX.sh (on Linux platforms). This procedure will remove all the additional files installed (including the local distribution of Conda). After this, just delete the folder containing the VEGA application.

To start the application, just move to the VEGA folder and run the file Vega-launcher-WIN.bat (on Windows platforms) or Vega-launcher-LINUX.sh (on Linux platforms).

If you experience problems with the above mentioned launchers, you can just run the file Vega-GUI-1.2.5.jar with the Java Runtime Environment (JRE). On most systems, it is enough to double-click it. If you are not able to directly run it , open a terminal window (like Command Prompt on Windows systems or BASH shell on Linux), move to the Vega folder and type:

java -jar Vega-GUI-1.2.5.jar

Note: Vega may encounter problems when processing large datasets and/or using a high number of models at the same time. You can modify and use the launcher files (Vega-launcher-WIN.bat and Vega-launcher-LINUX.sh) to assign more memory to the application. In the launcher file modify the parameter "-Xmx2048m", where 2048m means that Vega uses 2,048 MB (2 GB), setting a higher number (for instance -Xmx4096m for running Vega with 4 GB of memory).

NOTE FOR MAC USERS:
On MAC platforms, the system could block the application because it is does not have an Apple recognized developer. If this happen, you should go into the "Privacy & security" settings and allow the system to run VEGA.
Furthermore, some of the Python QSAR models could be unable to run, due to problems with some Python libraries not available for all Mac systems. In this case, you will just find a missing prediction and an error message (such as "unable to calculate CDDD descriptors") but you will be still able to run all other models.

3. How to Use

VEGA has a user-friendly interface, organized in three tabs corresponding to the steps needed to perform calculations. In order to run the models, all the parameters of the three tabs must be correctly set. In the first tab,INSERT, you have to build the compounds dataset. In the second tab, SELECT, you have to select the models to be calculated. In the third tab, EXPORT, you have to select the type and the destination folder for saving the generated reports.

3.1 Insert

In this tab, you have to insert one or more structure(s) in order to build the dataset that will be used for the calculations. You can insert the structure by directly typing it, using the SMILES format, in the “Load” text field; it is also possible to copy-and-paste single SMILES in the text field. Then, click the “Load” button, or just press enter to insert the structure. If the SMILES format of the structure is correctly typed, then it is imported in the table below the text field.

Also, you can import one or more structure(s) from SDF or SMILES file formats. SDF (structure-data file) is a chemical data file format developed by MDL. This kind of text-based file is specifically formatted to include multiple structures supplied with data (e.g., name, CAS, molecular mass, activity etc.). SMILES files are text files containing multiple SMILES strings (one molecule for each line of the file). For this kind of files, also tab-separated text files are accepted: for each line, other information for the given SMILES molecule (like its id or CAS number) can be reported, separated by a tab character.

To import a file, click the “Import file” button and select the file to be loaded. Depending on the type of the file, further information could be required:
- For SDF files, if some tags are available a form will give you the chance to choose one of the tags as the id (name) of the molecules that will be loaded.
- For SMILES files, if the file contains tab-separated fields a form will give you the chance to choose what field contains the SMILES string, and to optionally choose another field as the id (name) of the molecules that will be loaded.
In both cases, it will be possible to select the option for parsing the chosen molecule's id as a CAS number: the read string will be normalized so that CAS numbers written with separation characters other than “-”, or CAS written without any separation character will be reported in a default style (like 12345-67-8).

After successfully importing the structures using one or both of the ways above, you can visualize the structure by clicking on it.

By selecting some of the imported structure(s) and clicking the "Delete" button it is possible to erase them. To clean the whole table just click "Delete All" button.

3.2 Select

In the second tab, you have to select the models to be calculated. The currently available models are organized in six main categories that can be filtered:

- Human Toxicity
- EcoToxicity
- Fate & Distribution
- Physical-Chemical properties
- Human PBPK
- Ecological PBPK

For each category, all models available are shown grouped by end-point.

By clicking the question mark icon near each model, you can access to the details of the model and to the list of available files for that model: its user's guide (PDF document); the original dataset used for building the model (plain text file) containing the CAS number, the SMILES, the set (training or test set) and the experimental value for each compound; the QMRF document if available.

3.3 Export

In the third tab, you have to select the type of the reports to be generated, and the destination folder where the reports will be saved. Reports can be saved in two different formats, PDF and plain text. The supported output choices are the below:

- Multiple PDF documents – Results are saved in PDF format, one file for each calculated model

- PDF single document (ordered by model) – Results are saved in a single PDF report, ordered by model

- PDF single document (ordered by molecule) – Results are saved in a single PDF report, ordered by molecule

- Summary as tab-separated text file – Results are saved in a single text file, where for each model is reported only the final assessment
- Multiple tab-separated text files – Results are saved in text files, one for each model

For each choice, you have also to select the destination folder where the reports will be saved after the calculation. It is possible to select one or more export choices.

3.4 Predict

When the parameters of the three tabs described above are correctly filled, just click "Predict" button to start calculations. Please note that the calculation and report generation task may take long time, depending on the number of the imported substances and the selected models. In particular, note that the task of creating PDF reports can be particularly time-consuming when calculation are done for high numbers of compounds

4. Acknowledgments and contacts

VEGA application has been developed by Istituto di Ricerche Farmacologiche Mario Negri (Laboratory of Environmental Chemistry and Toxicology). The official website of the project is http://vega-qsar.eu

Please check each model's guide for acknowledgments on its development.

Please check the official website for acknowledgments on the EU projects that have supported the development of VEGA.

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